Prior-Anchored Deep Learning for Dense UAV Air-to-Ground Channel Knowledge Maps

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***Abstract*—Channel knowledge maps (CKMs) promise fast environment-aware radio planning, but dense air-to-ground CKM generation remains difficult when path loss, delay spread, and angular spread must all be predicted over unseen urban layouts. This paper condenses a full thesis project into its final technical line: a strict city-holdout evaluation protocol, train-only calibrated physical/statistical priors, and a bounded residual neural model with a Gaussian-mixture residual head. The final model predicts 513×513 dense maps from topology, geometric masks, UAV height, and frozen priors. On held-out cities it reaches 1.6519 dB path-loss RMSE, 26.5570 ns delay-spread RMSE, and 11.3854° angular-spread RMSE, improving the frozen priors for all targets. The result supports a hybrid design principle: use calibrated priors for stable radial and visibility-dependent structure, and reserve neural capacity for bounded local residuals and target-distribution mismatch.**

***Index Terms*—channel knowledge maps, UAV communications, radio-map prediction, path loss, delay spread, angular spread, Gaussian mixture models, physics-informed learning.**

I. Introduction

Channel knowledge maps (CKMs) store or predict radio-channel quantities as functions of position and environment, enabling faster planning and adaptation than repeated ray tracing or exhaustive measurements [1], [2]. In UAV-enabled 6G networks, CKMs are especially attractive because altitude changes both the geometry and the line-of-sight probability of the link [3], [4]. A practical CKM surrogate, however, must generalize to urban layouts not seen during training and must not hide hard propagation regimes behind a single global score.

The thesis behind this paper began as a generic dense image-to-image learning problem, but that formulation was not sufficient. Early U-Net and cGAN models could draw plausible maps, yet their physical errors remained large under a ground-only metric. PMNet/PMHHNet-style backbones improved smooth LoS structure but did not solve NLoS mode collapse or the spike-plus-tail nature of spread targets. The final design therefore moved away from "bigger backbone first" and toward a hybrid sequence:

1) define the evaluation contract before comparing models;
2) diagnose target distributions before choosing losses and heads;
3) calibrate strong train-only priors for predictable structure;
4) train a bounded residual model around those frozen priors.

The contribution is not a new propagation law or a claim of universal benchmark dominance. It is a defensible dense CKM recipe for one fixed UAV simulation contract: strict city holdout, valid-receiver masking, explicit LoS/NLoS and height reporting, and a final prior-anchored neural correction.

II. Related Work

Dense radio-map prediction has often been treated as image-to-image regression. RadioUNet [5] established convolutional radio-map estimation as a strong baseline; PMNet and the ICASSP path-loss challenge family further showed that U-Net-like architectures can perform well for path-loss maps [6], [7]. More recent models introduce attention, equivariance, or domain-shift machinery, including RMTransformer [8], RadioGUNet [9], PathFinder [10], and digital-twin radio-map generators such as AIRMap [11]. These papers provide the closest dense path-loss map context, but they generally do not match the present contract: continuous UAV height, strict city holdout on CKM samples, and three dense targets (path loss, delay spread, angular spread).

CKM work itself emphasizes that a radio map is not merely an image but a position-indexed channel side-information object [1], [2], [12]. This is why the evaluation protocol matters as much as the neural architecture. A random image split can reward memorization of urban morphology, whereas a city split tests whether the learned propagation surrogate transfers to unseen layouts.

Physics-informed and hybrid methods address a different weakness of pure image regression: the model should not need to rediscover stable propagation effects from pixels alone. ReVeal [13] uses physics-informed constraints for radio-environment mapping, while classical air-to-ground, COST/Hata, and two-ray models encode known distance, height, reflection, and elevation-angle structure [3], [14]–[19]. This paper uses those models only as shape priors; all deployment coefficients are calibrated from training cities under the same CKM masking convention used at inference.

The final neural head is also informed by distribution-aware wireless learning. Heteroscedastic regression [20], mixture-density wireless predictors [21]–[23], and recent stochastic radio-map models [24] all point to the same issue: NLoS and spread targets are not well represented by a single symmetric residual. For spread targets, the closest external references are often scalar channel-parameter predictors or standard channel-model statistics [18], [25]–[27], not dense maps, so they are not used as direct CKM benchmarks here.

III. Task and Evaluation Protocol

Each sample is a 513×513 urban topology map from the ray-traced CKM setting used throughout the thesis. The carrier

is fixed at 7.125 GHz, so frequency is not a learned variable; the main transmitter degree of freedom is height. There is one UAV transmitter at the map centre. The transmitter horizontal position is fixed, but its antenna height h_{tx} varies by sample and is treated as a central conditioning variable rather than as metadata. The spatial resolution is $1\,\mathrm{m}\times 1\,\mathrm{m}$ per pixel. Every non-building pixel is treated as an active receiver location, with the receiver antenna placed at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$ above the ground. Building pixels are excluded from the receiver set rather than treated as easy zero-error targets. The model predicts three aligned dense maps:

$$y_t \in \{\mathrm{PL}\ (\mathrm{dB}), \mathrm{DS}\ (\mathrm{ns}), \mathrm{AS}\ (\mathrm{deg})\}.$$

Let $m_i(p) = 1$ if pixel p of sample i is one of these valid ground receivers, and zero otherwise. For a split $\mathcal{S}$, the target-specific pixel-weighted RMSE is

$$\mathrm{RMSE}_t = \sqrt{\frac{\sum_{i\in\mathcal{S}}\sum_p m_i(p)\left(\hat{y}_{i,t}(p) - y_{i,t}(p)\right)^2}{\sum_{i\in\mathcal{S}}\sum_p m_i(p)}}. \quad (1)$$

The split is city-based rather than random over pixels or tiles. This matters: random splits would allow the model to see nearly identical urban morphology during training and testing. The final held-out test split contains 2590 samples from cities not used to optimize model weights. Validation is used for model selection; test is used only for the final generalization claim.

A. Quantized Targets and Continuous Outputs

The released CKM targets are not ideal real-valued measurements. Path loss is stored on an approximately 1 dB grid, while delay spread and angular spread are strongly integer-valued and visibly staircase-like in many maps. The model outputs are not quantized to these grids: the residual network emits continuous values, and all reported metrics compare those continuous outputs against the stored targets. This matters for interpretation. The final path-loss MAE is 0.9298 dB, close to the 1 dB label step, while the path-loss RMSE is 1.6519 dB because RMSE emphasizes the remaining hard tails. A cleaner or less quantized dataset would likely expose more of the continuous surrogate's resolution, especially for small residual improvements and spread-map boundaries.

IV. Method

A. Why the Final Method Is Hybrid

The development path exposed three distinct target regimes. First, LoS path loss contains stable radial, height-dependent, and coherent two-ray structure. Second, NLoS path loss is not just a smooth residual; it is a blocked-link distribution-placement problem. Third, delay spread and angular spread contain near-zero LoS spikes and NLoS-heavy tails. Thus a single point-regression CNN is asked to solve three different statistical problems at once.

The final method uses a division of labour. Frozen priors provide a strong mean prediction for structure that can be calibrated reliably from training data. The neural model receives those priors and learns bounded residual corrections. This makes the prior useful but not magical: distribution-first neural models showed that NLoS structure can be learned without explicit priors when the target distribution is represented correctly. The priors are reintroduced because they are cheaper, more stable, and more auditable for the predictable part of the map.

B. Frozen Priors

The path-loss prior combines a coherent two-ray LoS branch with a calibrated NLoS branch. The LoS side follows the textbook direct-plus-ground-reflected field structure [14], [15]; the NLoS side uses empirical urban and A2G attenuation shapes as a starting point [3], [16]–[18]. The UAV remains horizontally centred, so d_{2D} is the per-pixel distance from the centre transmitter to each receiver. The height h_{tx}, however, changes from sample to sample and is one of the main physical degrees of freedom in the method. For receiver height h_{rx} and carrier frequency f_{MHz}, the direct and reflected distances are

$$\begin{aligned} d_{\mathrm{LoS}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2}, \\ d_{\mathrm{ref}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2}. \end{aligned} \quad (2)$$

The LoS branch starts from

$$\mathrm{FSPL} = 32.45 + 20\log_{10}\left(\frac{d_{\mathrm{LoS}}}{1000}\right) + 20\log_{10}(f_{\mathrm{MHz}}) \quad (3)$$

and adds an effective coherent reflection correction

$$C_{2\mathrm{ray}} = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}}{d_{\mathrm{ref}}}e^{-j(k(d_{\mathrm{ref}}-d_{\mathrm{LoS}})+\phi(h_{\mathrm{tx}}))}\right|. \quad (4)$$

The final LoS prior is

$$\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{2\mathrm{ray}} + b(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}), \quad (5)$$

where b is a height-dependent bias and r is a smoothed radial residual fit. This LoS prior is not FSPL alone and it is not a learned linear regressor. The direct distance d_{LoS} sets the free-space baseline, the image-source distance d_{ref} supplies the ground-reflected path, and $k(d_{\mathrm{ref}} - d_{\mathrm{LoS}})$ is the extra phase accumulated by that reflected wave. The train-calibrated functions $\rho(h_{\mathrm{tx}})$ and $\phi(h_{\mathrm{tx}})$ are effective reflection amplitude and phase curves, fitted in 5 m UAV-height bins, smoothed across bins, and linearly interpolated to the actual sample height at inference. The terms $b(h_{\mathrm{tx}})$ and $r(d_{2D}, h_{\mathrm{tx}})$ are outside the textbook two-ray expression: they remove, respectively, a remaining height-dependent offset and a smooth range-dependent residual; these terms are also stored as height-conditioned calibration curves rather than as one global correction.

The LoS calibration is deliberately structured rather than fully flexible. It is fitted only on valid LoS ground pixels from training cities. First, the effective reflection amplitude ρ and phase ϕ are fitted in 5 m UAV-height bins to reproduce the ring-like constructive and destructive interference visible in LoS path-loss maps. Second, the per-height bias b removes a remaining vertical offset after the coherent two-ray fit. Third,

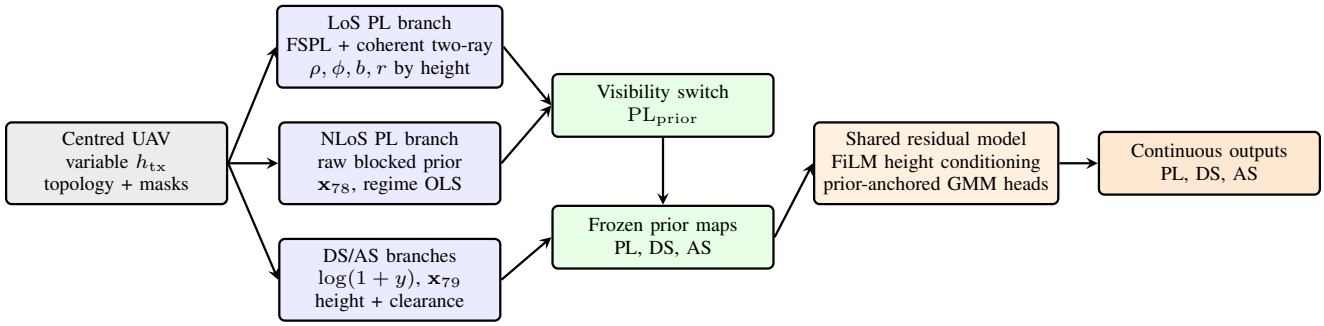

Fig. 1. Compact thesis-style pipeline. The transmitter is always centred, but its height varies by sample. Frozen height-aware priors are produced first and then used as channels for the final prior-anchored residual GMM model.

the radial residual table $r(d_{2D}, h_{\mathrm{tx}})$ averages the remaining error by distance from the centred transmitter, smooths it across radius, clips it to a small dB range, and stores it by height. At inference, the actual h_{tx} selects/interpolates these calibration curves, so the LoS prior changes continuously with antenna height even though the UAV horizontal coordinate remains fixed at the map centre.

The NLoS prior starts from the maximum of COST231-like and A2G-NLoS raw terms. The COST/Hata-shaped term is used as an urban-macro shape prior,

$$\begin{aligned} \mathrm{PL}_C &= 46.3 + 33.9\log_{10}(f_{\mathrm{MHz}}) \\ &\quad - 13.82\log_{10}(h_{\mathrm{tx}}) - a(h_{\mathrm{rx}}) \\ &\quad + \eta(h_{\mathrm{tx}})\log_{10}(d_{\mathrm{km}}) + 3, \\ \eta(h_{\mathrm{tx}}) &= 44.9 - 6.55\log_{10}(h_{\mathrm{tx}}), \end{aligned} \tag{6}$$

where C denotes the COST231-shaped branch, $d_{\mathrm{km}} = \max(d_{2D}/1000, 0.001)$, and $a(h_{\mathrm{rx}})$ is the usual mobile-height correction. The A2G envelope is

$$\begin{aligned} \mathrm{PL}_{\mathrm{A2G,NLoS}} &= \lambda_0(h_{\mathrm{tx}}, f) + \beta_{\mathrm{NLoS}} \\ &\quad + A_{\mathrm{NLoS}} \exp\left(-\frac{90-\theta^\circ}{\tau_{\mathrm{NLoS}}}\right), \end{aligned} \tag{7}$$

with $\lambda_0 = 20\log_{10}(4\pi h_{\mathrm{tx}} f/c)$. The raw blocked-link prior is the conservative envelope

$$P_0 = \mathrm{PL}_{\mathrm{NLoS,raw}} = \max(\mathrm{PL}_C, \mathrm{PL}_{\mathrm{A2G,NLoS}}). \tag{8}$$

This maximum is deliberately pessimistic: if the urban empirical curve predicts stronger attenuation it is kept, and if the elevation-aware A2G envelope is larger it is kept instead. The raw prior is then corrected by train-only regime-wise linear calibration:

$$\mathrm{PL}_{\mathrm{NLoS}} = \mathrm{clip}(\mathbf{x}_{78}^\top \mathbf{w}_{78,r}, 20, 180). \tag{9}$$

The path-loss prior is therefore not one formula with a small correction. It is two different calibrated feature extractors. The LoS side calibrates height-binned two-ray amplitude/phase, height bias, and radial residual curves. The NLoS side uses the raw blocked-link prior only as one feature in a 15-term morphology vector:

$$\begin{aligned} \mathbf{x}_{78} = [&P_0^2, P_0, \log(1+d_{2D}), \rho_{15}, \rho_{41}, h_{15}, h_{41}, \\ &\rho_{41}h_{41}, \rho_{41}\log(1+d_{2D}), n_{15}, n_{41}, s, \theta, n_{41}\theta, 1]^\top, \end{aligned} \tag{10}$$

where P_0 is the raw NLoS prior, ρ_{15} and ρ_{41} are the fractions of building pixels in 15×15 and 41×41 windows, h_{15} and h_{41} are local building-height summaries, n_{15} and n_{41} summarize local NLoS support, s is a shadow/blocker proxy, and θ is the normalized elevation-angle feature. The interaction $\rho_{41}h_{41}$ lets dense tall neighborhoods differ from sparse tall ones, $\rho_{41}\log(1+d_{2D})$ lets density matter differently with range, and $n_{41}\theta$ activates when obstruction and elevation jointly matter. The constant 1 is the regime-wise bias term. The deployed path-loss prior is selected by the geometric visibility mask:

$$\mathrm{PL}_{\mathrm{prior}} = m_{\mathrm{LoS}}\mathrm{PL}_{\mathrm{LoS}} + (1-m_{\mathrm{LoS}})\mathrm{PL}_{\mathrm{NLoS}}. \tag{11}$$

The regime index r is not a learned latent class. It is selected from observable scene descriptors: topology/morphology class, LoS/NLoS state, and antenna-height group, with fallback calibrations when an exact regime has too few training pixels. Thus the prior is auditable: one can inspect the raw blocked estimate P_0, the local morphology terms, the selected regime, and the final linear correction separately.

The delay-spread and angular-spread priors are fitted in a transformed domain,

$$z = \log(1+y), \qquad \hat{y} = \exp(\hat{z}) - 1, \tag{12}$$

because both standard channel models and the CKM histograms make spread quantities more regular in log space [18], [25]. The released spread maps have a near-zero mass plus a long tail: LoS direct-path pixels cluster near zero, while blocked or rooftop-scattered pixels create rare but important large values. Their raw prior is

$$\begin{aligned} z^{(0)} = &\log(1+b_q^{\mathrm{nat}}) + b_{\mathrm{top}} + a_1\log(1+d_{2D}) + a_2\theta_{\mathrm{inv}} \\ &+ a_3\rho_{41} + a_4h_{41} + a_5n_{41} + a_6n_{41}\theta_{\mathrm{inv}}, \end{aligned} \tag{13}$$

where q is the LoS/NLoS state. The native anchor b_q^{nat} gives a different starting scale to LoS and NLoS pixels, b_{top} shifts the baseline by topology class, $\log(1+d_{2D})$ captures range

TABLE I
TERM-BY-TERM INTERPRETATION OF THE MAIN FROZEN-PRIOR QUANTITIES.

Term	Prior block	Interpretation
d_{LoS}	LoS PL	3D direct transmitter–receiver distance; sets the FSPL baseline.
d_{ref}	LoS PL	Image-source reflected distance; gives the length and attenuation of the single ground-reflected ray.
$C_{2\mathrm{ray}}$	LoS PL	Coherent direct-plus-reflected correction; negative values mean constructive gain over FSPL and positive values mean fading loss.
$\rho(h_{\mathrm{tx}}), \phi(h_{\mathrm{tx}})$	LoS PL	Height-interpolated effective reflection amplitude and phase, calibrated from CKM rather than assumed from a fixed material.
$b(h_{\mathrm{tx}})$	LoS PL	Height-dependent dB offset remaining after the coherent two-ray fit.
$r(d_{2D}, h_{\mathrm{tx}})$	LoS PL	Smoothed radial residual profile; removes systematic range-dependent error left by the two-ray model.
P_0	NLoS PL	Raw blocked-link path-loss prior before morphology calibration.
ρ_k, h_k, n_k	NLoS PL and spreads	Local building density, local building height, and local NLoS support in a $k \times k$ window.
b_{41}, c_{41}, t_{41}	Spreads	Blocker depth, transmitter clearance, and fraction of buildings taller than the link-line proxy.
$\theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}$	Spreads	Elevation angle and its low-elevation complement; low elevation is where street-canyon and rooftop interactions are stronger.

growth, θ_{inv} increases the prior for low-elevation links, and ρ_{41}, h_{41}, and n_{41} summarize local building density, height, and NLoS support. This raw log prior is intentionally simple; it encodes the expected spike-plus-tail shape before CKM-specific calibration. A 23-term feature vector then augments $z^{(0)}$ with range, elevation, height, multi-scale morphology, blocker-depth, clearance, and interaction terms:

$$\begin{aligned}\mathbf{x}_{79} = [&(z^{(0)})^2, z^{(0)}, \log(1+d_{2D}), \theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}, h, h^2,\\ &\rho_{15}, \rho_{41}, h_{15}, h_{41}, n_{15}, n_{41}, n_{41}\log(1+d_{2D}),\\ &\rho_{41}\theta_{\mathrm{inv}}, b_{41}, 1, c_{41}, t_{41}, \theta_{\mathrm{norm}}\rho_{41},\\ &c_{41}\theta_{\mathrm{inv}}, t_{41}\rho_{41}, t_{41}n_{41}]^\top.\end{aligned} \tag{14}$$

Here h is normalized UAV height, b_{41} is a local blocker-depth summary, c_{41} is transmitter clearance over nearby buildings, and t_{41} is the fraction of locally taller buildings. These height terms are not incidental: they tell the spread prior whether the centred UAV is above the local skyline, near rooftop level, or below nearby high-rise clutter. The quadratic raw prior terms let the calibration bend the initial log-domain estimate, while the interaction terms represent cases where range, low elevation, dense buildings, and NLoS support become harmful together. The calibrated log prior is ridge-regularized per metric and regime:

$$\hat{\mathbf{w}}_{79,r} = \left(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{D}^\top\mathbf{D}\right)^{-1}\mathbf{X}^\top\mathbf{z}. \tag{15}$$

The calibrated log prior is then $\hat{z} = \mathbf{x}_{79}^\top\hat{\mathbf{w}}_{79,r}$. The diagonal regularizer $\mathbf{D}$ keeps the calibrated curve close to the raw prior unless the training residuals consistently justify a correction. In practice this matters for the spread targets: a flexible unregularized fit can chase rare high-spread pixels, whereas the ridge fit keeps the near-zero LoS bulk and the NLoS tail connected to the same physical features. All calibrations use training cities only. The spread-prior dictionary contains 114 fitted regime keys, but this number is not one Cartesian product. Per spread metric, the exact grid contributes 6 topology classes $\times$ 2 visibility states $\times$ 3 height bins $=$ 36 exact keys. The implementation then adds 12 height-collapsed topology/visibility fallbacks, 6 topology-wide fallbacks, 2 global visibility fallbacks, and 1 fully global fallback, for 57 keys per metric and 114 keys across DS and AS. Height is handled differently across the frozen estimators: the LoS PL branch interpolates its fitted 5 m height-bin curves, while the NLoS PL and spread branches use discrete antenna-height regimes with fallback keys; the spread model also keeps continuous height, clearance, and taller-building features inside $\mathbf{x}_{79}$. This heavy calibration is the point: the prior stores many small decisions about morphology, visibility, and height before the neural residual is trained. The same-protocol frozen-prior results are 1.7516 dB LoS PL, 3.3967 dB NLoS PL, 1.9246 dB overall PL, 28.10 ns DS, and 13.74° AS. The final neural model receives only the resulting prior maps as channels; the handcrafted features remain inside the frozen prior estimators rather than becoming additional learned inputs.

C. Prior-Anchored Residual Model

The neural model receives nine aligned channels: topology, LoS mask, NLoS mask, ground mask, combined path-loss prior, LoS path-loss prior, NLoS path-loss prior, delay-spread prior, and angular-spread prior. Height is encoded with sinusoidal features and injected through FiLM modulation [28], so the final network can adapt the residual correction continuously between the calibrated height regimes instead of learning one height-agnostic correction. The backbone is a shared U-Net-style encoder-decoder with GroupNorm, FiLM conditioning, and task-specific residual heads.

The height path mirrors the detailed thesis implementation. The scalar h_{tx} is expanded to a 32-dimensional sinusoidal code using 16 log-spaced frequencies over the UAV-height range,

$$\mathbf{e}(h_{\mathrm{tx}}) = \left[\sin\frac{h_{\mathrm{tx}}}{f_i}, \cos\frac{h_{\mathrm{tx}}}{f_i}\right]_{i=1}^{16} \in \mathbb{R}^{32}, \tag{16}$$

then mapped by an MLP to a $\mathbb{R}^{160}$ condition vector. Each FiLM site projects this vector to channel-wise scale and shift parameters and modulates a feature map as

$$\mathbf{X}' = \mathbf{X} \odot (1+\mathbf{g}) + \mathbf{b}. \tag{17}$$

Encoder FiLM uses the height condition directly; decoder FiLM and the global task heads use a fused condition formed from the height code and the pooled bottleneck context. Thus the same centred topology can receive different residual corrections when the UAV is at 30 m, 150 m, or 350 m.

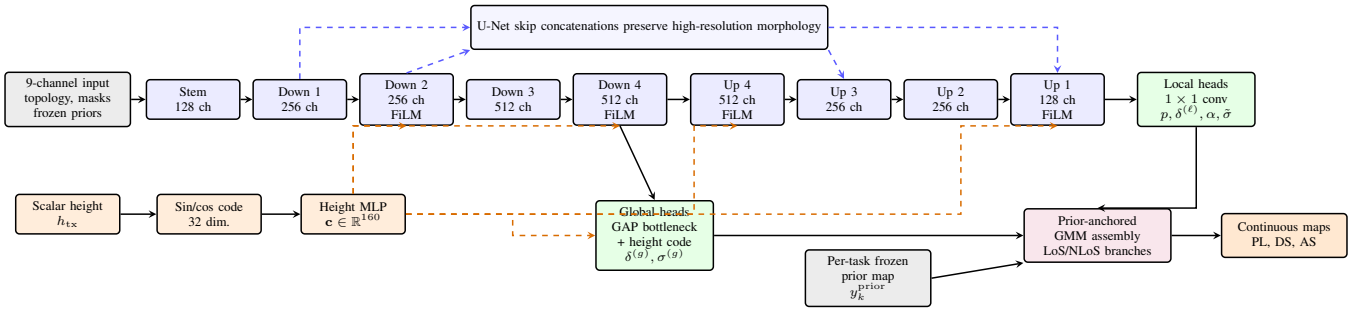

Fig. 2. Final shared deep-learning architecture adapted from the thesis diagram. The map stack carries topology, masks, and frozen priors; the scalar UAV height is encoded separately and injected through FiLM. A shared U-Net produces local residual fields, pooled bottleneck features produce global GMM parameters, and the prior-anchored GMM assembly emits continuous PL/DS/AS maps.

The head keeps the two-stage logic of the distribution-first predecessors, but moves it into residual space around the frozen prior. The global stage asks: "what residual distribution does this whole map, height, and topology imply?" The local stage asks: "where should each residual component be placed in the 513×513 map?" This is the key distinction from a direct U-Net regressor. The network does not directly overwrite the prior with a free-form map; it predicts a small residual distribution and then assigns that residual pixel by pixel.

For each task k, visibility region $r \in \{\mathrm{LoS}, \mathrm{NLoS}\}$, component $j \in \{1, 2, 3\}$, and pixel p, the global head receives the pooled bottleneck and fused height/context vector and outputs

$$\{\pi_{k,r,j}^{(g)}, \delta_{k,r,j}^{(g)}, \sigma_{k,r,j}^{(g)}\}_{j=1}^{3}.$$

After the implementation transforms, $\pi^{(g)}$ is a map-level residual mixture distribution, $\delta^{(g)}$ is a bounded global residual offset, and $\sigma^{(g)}$ is a bounded global uncertainty scale. The local head receives the full-resolution decoder feature map and outputs

$$\{p_{k,r,j}(p)\}_{j=1}^{3}, \quad \delta_{k,r}^{(\ell)}(p), \quad \alpha_{k,r}(p), \quad \tilde{\sigma}_{k,r}(p).$$

Here $p_{k,r,j}(p)$ is the per-pixel component assignment, the local residual $\delta^{(\ell)}$ shifts all components at that pixel, α is an anchor gate initialized with bias -2 so the model starts close to the prior, and $\tilde{\sigma}$ is the local uncertainty term. This extends the distribution-aware line of work from scalar mixture predictors to dense residual maps [21]–[23]. The native residual is clipped by target and visibility:

$$\Delta_{k,r,j}(p) = \mathrm{clip}\left(\delta_{k,r,j}^{(g)} + \delta_{k,r}^{(\ell)}(p), -\delta_{\max,k,r}, \delta_{\max,k,r}\right), \tag{18}$$

with caps of $7/25$ dB for PL LoS/NLoS, $20/20$ ns for DS, and $17^\circ/15^\circ$ for AS. The component mean is anchored to the frozen prior:

$$\begin{aligned} \mu_{k,r,j}^{\mathrm{native}}(p) &= y_k^{\mathrm{prior}}(p) + \alpha_{k,r}(p)\Delta_{k,r,j}(p), \\ 0 &\le \alpha_{k,r}(p) \le 1. \end{aligned} \tag{19}$$

PL is modeled in native dB space, while DS and AS use the log transform in Eq. (12). The deterministic map used for RMSE is the mixture expectation:

$$\hat{z}_{k,r}(p) = \sum_{j=1}^{3} p_{k,r,j}(p)\mu_{k,r,j}(p). \tag{20}$$

For PL, $\hat{y}_{k,r} = \hat{z}_{k,r}$. For DS and AS, the output is mapped back with $\hat{y}_{k,r} = \exp(\hat{z}_{k,r}) - 1$. The final map selects the LoS or NLoS branch with the explicit geometric mask. The mixture variances combine map-level and local uncertainty,

$$\sigma_{k,r,j}(p) = \sqrt{\left(\sigma_{k,r,j}^{(g)}\right)^2 + (\tilde{\sigma}_{k,r}(p))^2}, \tag{21}$$

with both terms constrained to $[0.05, 3.0]$. During the residual/GMM stage these variances and mixture weights define a pixel-wise likelihood; during the final fine-tuning stage the same head remains in place but the reported output is the deterministic expectation above.

The two mixture-weight objects have different jobs. The global weights $\pi_{k,r,j}^{(g)}$ describe the sample-level residual histogram and are used by the distributional regularizer. The local weights $p_{k,r,j}(p)$ are the pixel-wise assignment probabilities used in the final dense map. During the residual/GMM stage the per-pixel density is

$$q_{k,r}(z_k(p) \mid p) = \sum_{j=1}^{3} p_{k,r,j}(p)\,\mathcal{N}\big(z_k(p); \mu_{k,r,j}(p), \sigma_{k,r,j}^2(p)\big), \tag{22}$$

and the regional NLL is the masked average of $-\log q_{k,r}(z_k(p) \mid p)$ over valid LoS or NLoS pixels. Separately, the KL term bins the true residuals $\epsilon_k(p) = y_k(p) - y_k^{\mathrm{prior}}(p)$ into a soft empirical histogram and compares it with the histogram implied by $\{\pi_{k,r,j}^{(g)}, \delta_{k,r,j}^{(g)}\}$. This is what makes the head "distribution-first": before the decoder places values pixel by pixel, the global branch is pressured to put mixture mass where the map residuals actually live.

The final prior-anchored model should not be read as only the final fine-tuning loss. It has two active training phases: a large-capacity residual/GMM stage with probabilistic losses

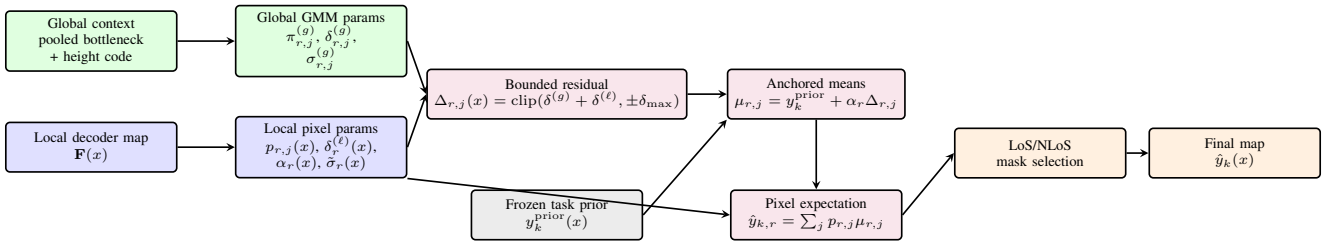

Fig. 3. Final prior-anchored GMM head. The global branch predicts the sample-level residual mixture; the local branch places that distribution pixel by pixel through component weights, residual refinements, gates and local uncertainty. Component means are anchored to the frozen prior and the deterministic map is the pixel-wise mixture expectation after LoS/NLoS selection.

enabled, followed by an RMSE/MAE-dominant fine-tuning stage. The composite objective implemented by the model is

$$\mathcal{L} = \sum_k w_k \big(\lambda_{\mathrm{NLL}}\mathcal{L}_{\mathrm{NLL},k} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL},k} + \lambda_{\mathrm{mom}}\mathcal{L}_{\mathrm{mom},k} + \lambda_{\mathrm{anchor}}\mathcal{L}_{\mathrm{anchor},k} + \lambda_{\mathrm{guard}}\mathcal{L}_{\mathrm{guard},k} + \lambda_{\mathrm{out}}\mathcal{L}_{\mathrm{out},k} + \lambda_{\mathrm{RMSE}}\mathcal{L}_{\mathrm{RMSE},k} + \lambda_{\mathrm{MAE}}\mathcal{L}_{\mathrm{MAE},k}\big). \tag{23}$$

The probabilistic terms matter during the residual/GMM stage. The NLL term teaches the head to place mixture mass around the transformed target, the KL and moment terms keep the mixture distribution from becoming a numerically convenient but physically uninformative density, and the outlier-budget term discourages a solution that spends most of its capacity on the easy bulk while leaving large tails unconstrained. The final fine-tuning stage then disables those terms and optimizes the deterministic mixture expectation for RMSE/MAE, with a small anchor and guard that keep the correction close to the prior. Table II summarizes the roles. The selected checkpoint is therefore reported as a deterministic RMSE improvement over the frozen priors, not as a calibrated probabilistic predictor.

V. Experimental Results

A. Main Test Results

Table V reports the final unseen-city test metrics. The model improves the frozen prior for all three targets. Path loss is the cleanest success case because both RMSE and MAE improve. Delay spread improves RMSE while MAE slightly increases; this is consistent with reducing rare high-delay errors while introducing small local shifts. Angular spread improves RMSE strongly and is approximately neutral in MAE.

The original numerical targets were 5 dB, 50 ns, and 20°. The final model clears all three on unseen cities. More importantly, it does so jointly: the same network predicts PL/DS/AS instead of using separate specialist models.

B. LoS/NLoS and Split Diagnostics

Table VI separates valid pixels by the geometric visibility mask. The residual model improves the prior in both LoS and NLoS pixels for all targets. This is important because global RMSE could otherwise hide a failure in the physically hard NLoS subset.

The validation/test sanity check is also useful. Validation stores the model selection values; the test row is the real generalization result. Delay spread worsens more on test because the held-out city mix contains harder high-delay cases. The frozen delay-spread prior shows the same validation-to-test worsening, confirming that this is a city-mix effect rather than an evaluator change.

C. Prior and Split Breakdowns

The frozen priors are strong enough to deserve their own reporting. They are not simply baselines for the final neural model; they encode most of the stable geometry and therefore determine what residual problem the network sees. Table VIII shows the height dependence of the calibrated path-loss prior. This is not post-hoc stratification: UAV height is part of the estimator itself through the centred geometry, elevation angle, height-binned calibration, and final FiLM conditioning. The trend is strongest in NLoS: the low-altitude bin remains the hardest because many receivers are blocked by local urban structure, while high UAVs see fewer deep shadow pockets. Table IX shows the matching spread-prior pattern. Delay spread in particular collapses with altitude, from 39.90 ns overall in the lowest bin to 3.70 ns above 300 m.

The path-loss prior also varies by topology. Its overall RMSE ranges from 1.43 dB in open sparse low-rise maps to 2.97 dB in dense block high-rise maps. The NLoS branch is hardest in sparse vertical and dense high-rise regimes, reaching 3.91 dB and 4.09 dB; the overall pixel-weighted prior remains 1.92 dB.

The final residual model improves the priors consistently across data splits, not only on the validation split used for model selection. Table X reports the model-minus-prior RMSE deltas. The test row is the main generalization claim. The all-city row is not used for model selection, but it checks that the gain is not a held-out split artifact.

The all-city subexpert audit groups samples by morphology (`open`, `mixed`, `dense`) and antenna-height bin. It is a stricter check than the global row because it asks whether the residual model helps across the deployed prior regimes rather than only in one easy subset. Table XI reports RMSE deltas for the nine groups. All entries are negative, so the residual correction improves every morphology and height group for all three targets.

TABLE II
LOSS TERMS AVAILABLE IN THE FINAL PRIOR-ANCHORED OBJECTIVE AND THEIR PRACTICAL PURPOSE.

Term	Active stage	Role
GMM NLL	residual/GMM stage	Fits the per-pixel mixture density in native PL space and log-transformed spread space.
Distribution KL	residual/GMM stage	Regularizes the learned mixture so it remains close to the intended distributional structure instead of using degenerate weights or variances.
Moment matching	residual/GMM stage	Encourages the expectation and spread of the mixture to agree with target moments, stabilizing deterministic extraction from the GMM.
Outlier budget	residual/GMM stage	Keeps rare large residuals visible in the objective instead of letting the global average hide them.
Anchor	both stages	Penalizes unnecessary deviation from the frozen prior; this is especially important because the prior is already a strong calibrated predictor.
Prior guard	both stages	Adds a penalty when the model becomes worse than the prior at valid pixels, discouraging destructive corrections.
RMSE / MAE	both stages, dominant in final fine-tuning	Optimizes the deterministic map reported in the final evaluation; RMSE handles tails and MAE tracks typical absolute error.

TABLE III
TRAINING CONFIGURATION OF THE FINAL LINE.

Item	Value
Image size	513×513
Input channels	9 map channels plus height FiLM conditioning
Backbone	GroupNorm U-Net, base width 128, FiLM dim. 160
Residual head	3 mixture components, bounded residuals
Optimizer	AdamW, learning rate 5×10^{-5}, weight decay 0.01
Batching	batch size 1, gradient accumulation 8
Loss schedule	Large-capacity residual/GMM stage plus final residual fine-tuning; see Table IV

TABLE IV
FINAL PRIOR-ANCHORED LOSS SCHEDULE. THE NON-FINE-TUNED RESIDUAL/GMM STAGE IS PART OF THE FINAL PIPELINE, EVEN THOUGH THE REPORTED CHECKPOINT IS SELECTED AFTER THE DETERMINISTIC FINE-TUNING STAGE.

Setting	Residual/GMM stage	Final fine-tuning
NLL / KL / moment / outlier	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0
RMSE / MAE	1.50 / 0.10	1.00 / 0.05
Anchor / guard	0.05 / 0.10	0.01 / 0.02
Task weights	PL 1.0, DS 1.0, AS 1.0	PL 1.0, DS 0.25, AS 0.25
Learning rate	2×10^{-4}	5×10^{-5}

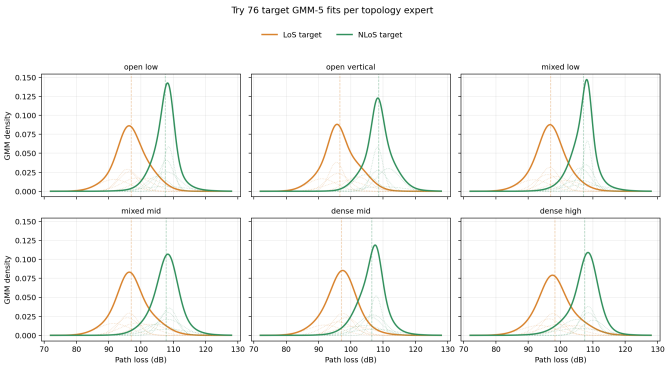

Fig. 4. Path-loss target GMM fits by topology expert. The diagnostic exposed different LoS/NLoS marginals and motivated distribution-aware heads before returning to prior-anchored residual learning.

D. Why Distribution Diagnostics Matter

Fig. 4 shows the diagnostic that changed the modeling strategy: LoS and NLoS path loss do not form a single easy Gaussian map target. The distribution-first predecessor modeled the marginal with GMM heads and showed that the NLoS collapse was not inevitable. The final model keeps this lesson but anchors the residual around explicit priors.

The internal diagnostic numbers are kept as scale references, not as final claims. The distribution-first path-loss GMM experts reached 4.43 dB LoS and 3.34 dB NLoS pixel-aggregate RMSE, showing that NLoS collapse was a modeling choice rather than a necessity. The distribution-first spread experts reached 30.96 ns delay-spread RMSE and 11.78° angular-spread RMSE, motivating log-domain and mixture-aware treatment of spread targets. The frozen priors then provided stronger anchors: 1.9246 dB path-loss RMSE, 28.10 ns delay-spread RMSE, and 13.74° angular-spread RMSE.

For spread targets, quantized labels make visual and RMSE interpretation harder. The released delay and angular maps are integer-valued; smooth predictions can be penalized across staircase-like target boundaries even when the morphology is plausible. This does not invalidate RMSE, but it warns against overinterpreting small local residuals.

E. External Metric Context

Table XII places the final path-loss result beside the closest available path-loss-side references. The table is deliberately compact and cautious: datasets, height assumptions, target definitions, and splits differ. No scalar indoor or link-level spread predictor is used as a direct dense-map leaderboard.

The fair conclusion is narrow but useful. The final path-loss number is in the same band as strong dense radio-map references, while the spread results should be interpreted primarily against same-protocol internal baselines because no external dense CKM PL/DS/AS benchmark was found.

F. Qualitative Residual Behavior

Fig. 5 is the qualitative check. The model receives the frozen prior and predicts a correction. The useful visual evidence is not merely that the prediction resembles the target; it is that the correction row tends to anti-correlate with the prior-error row, which is the expected behavior of a prior-anchored residual corrector.

TABLE V
FINAL MODEL TEST RESULTS ON UNSEEN CITIES. NEGATIVE Δ MEANS THAT THE RESIDUAL MODEL IMPROVES THE FROZEN PRIOR.

Target	Model RMSE	Prior RMSE	ΔRMSE	Model MAE	Prior MAE	ΔMAE	Unit
Path loss	1.6519	1.9383	-0.2865	0.9298	1.2319	-0.3020	dB
Delay spread	26.5570	28.1023	-1.5453	7.9897	7.3033	+0.6864	ns
Angular spread	11.3854	13.7416	-2.3562	5.0088	4.9890	+0.0198	deg

TABLE VI
FINAL TEST RMSE SEPARATED BY LOS AND NLOS PIXELS.

Target	Scope	Model	Prior	Δ
PL	LoS	1.4675	1.7519	-0.2844
PL	NLoS	3.1482	3.5143	-0.3661
DS	LoS	25.4419	27.4851	-2.0432
DS	NLoS	29.2045	29.6157	-0.4112
AS	LoS	12.3510	15.2818	-2.9308
AS	NLoS	8.4424	8.6614	-0.2191

TABLE VII
VALIDATION VERSUS HELD-OUT TEST CHECK FOR THE SAME SELECTED MODEL.

Split	Samples	PL RMSE	DS RMSE	AS RMSE
Validation	2750	1.6157	22.1061	11.3782
Test	2590	1.6519	26.5570	11.3854

TABLE VIII
CALIBRATED PATH-LOSS PRIOR RMSE BY UAV HEIGHT.

Height bin	Overall	LoS	NLoS	Samples
12–50 m	2.180	1.797	3.756	1211
50–150 m	1.936	1.785	3.228	2634
150–300 m	1.680	1.666	2.259	786
300–500 m	1.627	1.619	2.255	362

TABLE IX
CALIBRATED SPREAD-PRIOR RMSE BY UAV HEIGHT.

Height bin	Metric	Overall	LoS	NLoS
12–50 m	DS	39.90	43.04	35.13
50–150 m	DS	27.17	26.72	28.29
150–300 m	DS	8.70	8.79	8.13
300–500 m	DS	3.70	3.80	2.76
12–50 m	AS	15.69	18.81	9.93
50–150 m	AS	14.19	15.89	8.41
150–300 m	AS	10.39	11.07	4.49
300–500 m	AS	8.85	9.34	2.85

TABLE X
RMSE IMPROVEMENT OVER FROZEN PRIORS BY SPLIT. NEGATIVE VALUES MEAN THE RESIDUAL MODEL IMPROVES THE PRIOR.

Split	Samples	PL	DS	AS
Train	10840	-0.2895	-2.0157	-2.3578
Validation	2750	-0.2758	-1.3433	-2.4396
Test	2590	-0.2865	-1.5453	-2.3562
All	16180	-0.2867	-1.8406	-2.3710

G. Runtime

The final generator was compared against the supplied MATLAB ray-tracing path on Barcelona layouts at full 513×513 resolution on the same laptop: an AMD Ryzen 5 5600H CPU and an NVIDIA RTX 3050 Ti Laptop GPU. The surrogate includes LoS/NLoS mask generation, frozen priors, final neural prediction, and default numeric export after the checkpoint is loaded once. It runs in 0.88 s per sample versus 99.89 s for the local ray-tracing path, a $113.7\times$ speed-up under this measurement setup. The ray-tracing row uses the CPU path; the surrogate row uses the laptop GPU with mixed precision.

TABLE XI
ALL-CITY FINAL-MODEL RMSE DELTAS BY MORPHOLOGY/HEIGHT SUBEXPERT. NEGATIVE VALUES MEAN THE FINAL MODEL IMPROVES THE FROZEN PRIOR.

Group	Samples	PL	DS	AS
`dense/high`	1957	-0.3602	-0.6473	-2.6405
`dense/low`	2018	-0.3359	-1.5779	-2.3994
`dense/mid`	2055	-0.3288	-1.4074	-2.8587
`mixed/high`	2156	-0.2931	-0.6794	-2.3186
`mixed/low`	2060	-0.2745	-2.8379	-2.5845
`mixed/mid`	2094	-0.2781	-2.0522	-2.8200
`open/high`	1272	-0.2721	-0.5428	-1.2874
`open/low`	1318	-0.2363	-3.1170	-2.0271
`open/mid`	1250	-0.2941	-1.5789	-1.7804

TABLE XII
COMPACT COMPARISON WITH NEARBY PATH-LOSS-SIDE METRICS.

Reference	PL-side metric	Reading / caveat
RadioUNet [5]	1.6 dB–3.1 dB equivalent PL RMSE	Final PL is in this band; no UAV height or spread targets.
RadioGUNet [9]	1.304 dB DPM; 1.936 dB IRT	Final PL is between those values; outdoor RadioMapSeer setting.
PMNet / ICASSP [6], [7]	1.38 dB converted score	Final PL is $1.20\times$ worse; scale context only, no joint spread prediction.
ReVeal [13]	1.95 dB outdoor sparse-measurement RMSE	Final PL is $1.18\times$ better; sparse mapping rather than dense generation.
RMTransformer / PathFinder [8], [10]	≈ 1.82 dB / ≈ 1.19 dB converted references	Useful scale checks, but random/specialized splits and different assumptions.
AIRMap [11]	Path-gain error < 4 dB against ray tracing	Same broad digital-twin motivation, but path gain and 200×200 maps.
Final model (ours)	**1.6519 dB PL; 26.5570 ns DS;** 11.3854° **AS**	**Strict CKM city holdout, centred variable-height UAV, dense** 513×513 **PL/DS/AS.**

VI. DISCUSSION

The paper version reveals what the full thesis should preserve. The core argument is not the 80-attempt chronology. It is the chain: strict evaluation contract, distribution diagnosis, calibrated priors, bounded residual correction, and held-out-city evidence. Most development history is valuable only when it supports one of those links.

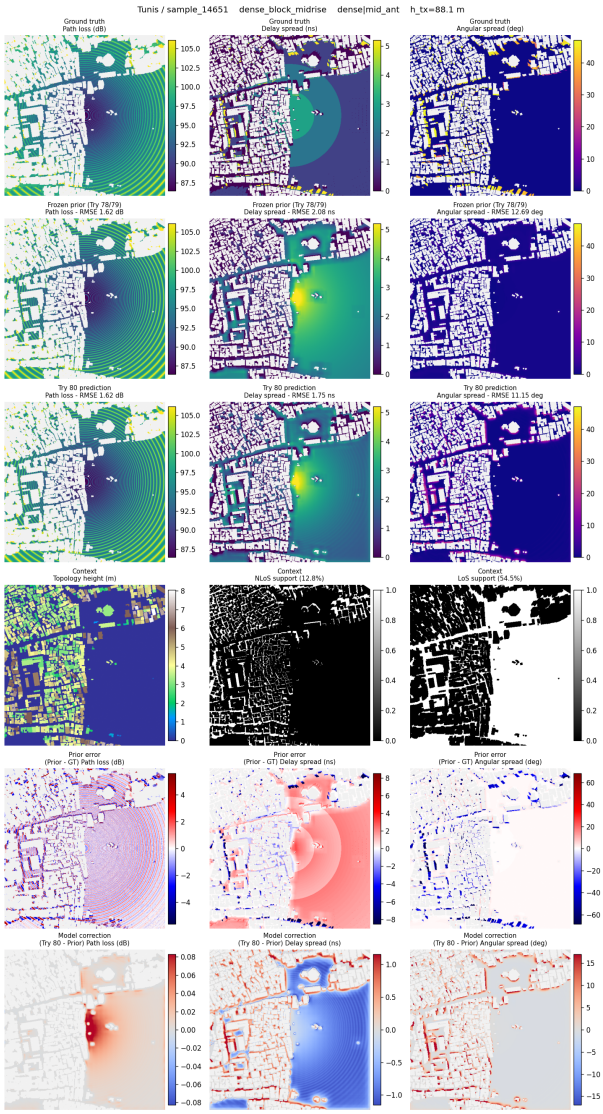

Fig. 5. Representative final-model inspection panel for a dense, mid-antenna validation sample in Tunis. Columns are path loss, delay spread, and angular spread. Rows show ground truth, frozen prior, final prediction, context/support, prior error, and model correction.

TABLE XIII
RUNTIME COMPARISON FOR ONE FULL-RESOLUTION BARCELONA CKM SAMPLE ON THE SAME RYZEN 5 5600H / RTX 3050 TI LAPTOP HARDWARE.

Method	Timed path	Time
MATLAB RT	Pixel receivers outside buildings; PL/DS/AS parsed from paths; CPU path	99.89 s
Final generator	Masks, priors, PL/DS/AS prediction, numeric export; CUDA mixed precision	0.88 s

Several limitations remain. First, the labels come from ray tracing rather than field measurements, so the result is simulator agreement, not over-the-air validation. Second, target quantization limits how much sub-unit error structure can be learned. This is not a limitation of the output layer: the model outputs continuous PL/DS/AS values. It is a limitation of the stored labels. For path loss, a 0.9298 dB final MAE against labels quantized at about 1 dB is already close to the label step, even though the RMSE still reveals harder tails. Third, the final model is selected under one city-holdout split and one final training seed. Fourth, the dataset fixes frequency, antenna pattern, and simulator assumptions; deployment would require conditioning or calibration across those variables.

VII. CONCLUSION

A dense UAV CKM predictor should not be designed as a generic image generator. In this setting, the best final result came from a hybrid prior-anchored formulation: frozen calibrated priors model stable radial, visibility, and spread-statistical structure; a shared neural model learns bounded residuals for the remaining local errors. The final model improves the frozen priors on unseen cities for path loss, delay spread, and angular spread, and produces a fast full-resolution generator. As a compression of the thesis, this paper suggests

that the full document can be shortened aggressively by keeping the final design chain and moving most chronological attempt detail to appendices or repository documentation.

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